Laboratório 04

ra: 175886

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Q1. Carregue os pacotes readxl e tidyverse.

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

Q2. Crie uma variável chamada fname que contenha o caminho para o arquivo WDIEXCEL.xlsx, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando. file.exists

```
fname <-  "\\\\smb\\ra175886\\WindowsDesktop\\ME315 B\\Lab 4\\WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

Q3. Leia a planilha "Country" e armazene no objeto countries.

```r
countries <- read_excel(fname, sheet = "Country")
countries
```

```
# A tibble: 264 x 31
   `Country Code` `Short Name`           `Table Name` `Long Name` `2-alpha code`
   <chr>          <chr>                  <chr>        <chr>       <chr>
 1 ABW            Aruba                  Aruba        Aruba       AW
 2 AFE            Africa Eastern and So~ Africa East~ Africa Eas~ ZH
 3 AFG            Afghanistan            Afghanistan  Islamic St~ AF
 4 AFW            Africa Western and Ce~ Africa West~ Africa Wes~ ZI
 5 AGO            Angola                 Angola       People's R~ AO
 6 ALB            Albania                Albania      Republic o~ AL
 7 AND            Andorra                Andorra      Principali~ AD
 8 ARB            Arab World             Arab World   Arab World  1A
 9 ARE            United Arab Emirates   United Arab~ United Ara~ AE
10 ARG            Argentina              Argentina    Argentine ~ AR
# i 254 more rows
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

Q4. Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha).

```r
countries <- countries %>%
  filter(`Country Code` %in% c("BRA", "ARG", "CHL", "URY", "COL"))
countries
```

```
# A tibble: 5 x 31
  `Country Code` `Short Name` `Table Name` `Long Name`            `2-alpha code`
  <chr>          <chr>        <chr>        <chr>                  <chr>
1 ARG            Argentina    Argentina    Argentine Republic     AR
2 BRA            Brazil       Brazil       Federative Republic o~ BR
3 CHL            Chile        Chile        Republic of Chile      CL
4 COL            Colombia     Colombia     Republic of Colombia   CO
5 URY            Uruguay      Uruguay      Oriental Republic of ~ UY
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
```

```
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

Q5. Leia a planilha "Data" e armazene no objeto WDI.

```r
WDI <- read_excel(fname, sheet = "Data")
WDI
```

```
# A tibble: 396,970 x 70
   `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
   <chr>          <chr>          <chr>            <chr>             <dbl>  <dbl>
 1 Africa Easter~ AFE            Access to clean~ EG.CFT.ACCS.ZS       NA     NA
 2 Africa Easter~ AFE            Access to clean~ EG.CFT.ACCS.RU.~     NA     NA
 3 Africa Easter~ AFE            Access to clean~ EG.CFT.ACCS.UR.~     NA     NA
 4 Africa Easter~ AFE            Access to elect~ EG.ELC.ACCS.ZS       NA     NA
 5 Africa Easter~ AFE            Access to elect~ EG.ELC.ACCS.RU.~     NA     NA
 6 Africa Easter~ AFE            Access to elect~ EG.ELC.ACCS.UR.~     NA     NA
 7 Africa Easter~ AFE            Account ownersh~ FX.OWN.TOTL.ZS       NA     NA
 8 Africa Easter~ AFE            Account ownersh~ FX.OWN.TOTL.FE.~     NA     NA
 9 Africa Easter~ AFE            Account ownersh~ FX.OWN.TOTL.MA.~     NA     NA
10 Africa Easter~ AFE            Account ownersh~ FX.OWN.TOTL.OL.~     NA     NA
# i 396,960 more rows
# i 64 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
#   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
#   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
#   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
#   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>, ...
```

Q6. Selecione apenas as colunas:

- Country Code
- Indicator Code
- E todos os anos de 1960-2018

```r
WDI <- WDI %>%
  select(`Country Code`, `Indicator Code`, all_of(as.character(1960:2018)))
WDI
```

```
# A tibble: 396,970 x 61
   `Country Code` `Indicator Code`  `1960` `1961` `1962` `1963` `1964` `1965`
   <chr>          <chr>              <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
 1 AFE            EG.CFT.ACCS.ZS        NA     NA     NA     NA     NA     NA
 2 AFE            EG.CFT.ACCS.RU.ZS     NA     NA     NA     NA     NA     NA
 3 AFE            EG.CFT.ACCS.UR.ZS     NA     NA     NA     NA     NA     NA
 4 AFE            EG.ELC.ACCS.ZS        NA     NA     NA     NA     NA     NA
 5 AFE            EG.ELC.ACCS.RU.ZS     NA     NA     NA     NA     NA     NA
 6 AFE            EG.ELC.ACCS.UR.ZS     NA     NA     NA     NA     NA     NA
 7 AFE            FX.OWN.TOTL.ZS        NA     NA     NA     NA     NA     NA
 8 AFE            FX.OWN.TOTL.FE.ZS     NA     NA     NA     NA     NA     NA
 9 AFE            FX.OWN.TOTL.MA.ZS     NA     NA     NA     NA     NA     NA
10 AFE            FX.OWN.TOTL.OL.ZS     NA     NA     NA     NA     NA     NA
# i 396,960 more rows
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
#   `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
#   `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
#   `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, ...
```

Q7. Confirme se os dados wm WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

```r
WDI <- WDI %>%
  # 1. Transforma os anos de colunas para linhas (exclui as colunas de código)
  pivot_longer(
    cols = -c(`Country Code`, `Indicator Code`),
    names_to = "YEAR",
    values_to = "value"
  ) %>%
  # 2. Transforma os códigos dos indicadores em colunas
  pivot_wider(
    names_from = `Indicator Code`,
    values_from = value
  )
WDI
```

```
# A tibble: 15,635 x 1,500
   `Country Code` YEAR  EG.CFT.ACCS.ZS EG.CFT.ACCS.RU.ZS EG.CFT.ACCS.UR.ZS
   <chr>          <chr>          <dbl>             <dbl>             <dbl>
 1 AFE            1960              NA                NA                NA
```

```
 2 AFE              1961              NA                NA                NA
 3 AFE              1962              NA                NA                NA
 4 AFE              1963              NA                NA                NA
 5 AFE              1964              NA                NA                NA
 6 AFE              1965              NA                NA                NA
 7 AFE              1966              NA                NA                NA
 8 AFE              1967              NA                NA                NA
 9 AFE              1968              NA                NA                NA
10 AFE              1969              NA                NA                NA
# i 15,625 more rows
# i 1,495 more variables: EG.ELC.ACCS.ZS <dbl>, EG.ELC.ACCS.RU.ZS <dbl>,
#   EG.ELC.ACCS.UR.ZS <dbl>, FX.OWN.TOTL.ZS <dbl>, FX.OWN.TOTL.FE.ZS <dbl>,
#   FX.OWN.TOTL.MA.ZS <dbl>, FX.OWN.TOTL.OL.ZS <dbl>, FX.OWN.TOTL.40.ZS <dbl>,
#   FX.OWN.TOTL.PL.ZS <dbl>, FX.OWN.TOTL.60.ZS <dbl>, FX.OWN.TOTL.SO.ZS <dbl>,
#   FX.OWN.TOTL.YG.ZS <dbl>, per_si_allsi.adq_pop_tot <dbl>,
#   per_allsp.adq_pop_tot <dbl>, per_sa_allsa.adq_pop_tot <dbl>, ...
```

Q8. Confirme se a coluna referente ao ano está em formato inteiro.

```r
WDI <- WDI %>%
  mutate(YEAR = as.integer(YEAR))

class(WDI$YEAR)
```

```
[1] "integer"
```

Q9. Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas:

- Country Code
- Short Name
- YEAR
- SP.DYN.CBRT.IN

```r
WDI_filtrado <- WDI %>%
  inner_join(countries, by = "Country Code") %>%
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT.IN)
WDI_filtrado
```

```
# A tibble: 295 x 4
   `Country Code` `Short Name`  YEAR SP.DYN.CBRT.IN
```

```
   <chr>          <chr>        <int>          <dbl>
 1 ARG            Argentina     1960           24.2
 2 ARG            Argentina     1961           23.9
 3 ARG            Argentina     1962           23.8
 4 ARG            Argentina     1963           23.5
 5 ARG            Argentina     1964           23.2
 6 ARG            Argentina     1965           22.6
 7 ARG            Argentina     1966           22.3
 8 ARG            Argentina     1967           22.4
 9 ARG            Argentina     1968           22.7
10 ARG            Argentina     1969           23.0
# i 285 more rows
```

Q10. Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

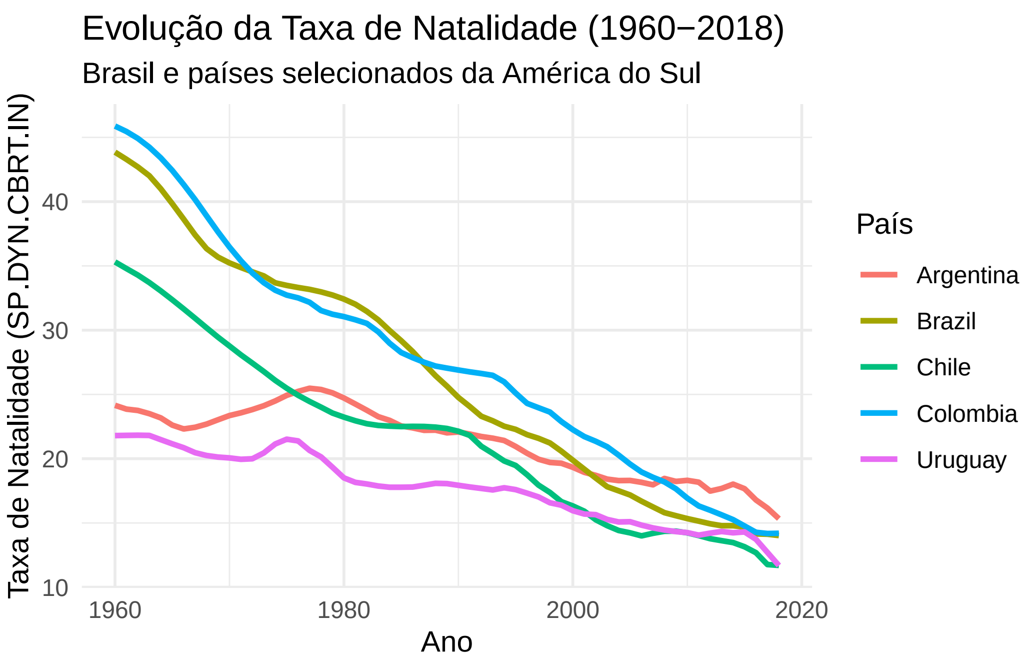
```r
WDI_filtrado <- WDI_filtrado %>%
  arrange(`Country Code`, YEAR)
WDI_filtrado
```

```
# A tibble: 295 x 4
   `Country Code` `Short Name`  YEAR SP.DYN.CBRT.IN
   <chr>          <chr>        <int>          <dbl>
 1 ARG            Argentina     1960           24.2
 2 ARG            Argentina     1961           23.9
 3 ARG            Argentina     1962           23.8
 4 ARG            Argentina     1963           23.5
 5 ARG            Argentina     1964           23.2
 6 ARG            Argentina     1965           22.6
 7 ARG            Argentina     1966           22.3
 8 ARG            Argentina     1967           22.4
 9 ARG            Argentina     1968           22.7
10 ARG            Argentina     1969           23.0
# i 285 more rows
```

Q11. Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```r
grafico <- ggplot(WDI_filtrado, aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +
  geom_line(linewidth = 1) +
  labs(
    title = "Evolução da Taxa de Natalidade (1960-2018)",
    subtitle = "Brasil e países selecionados da América do Sul",
    x = "Ano",
    y = "Taxa de Natalidade (SP.DYN.CBRT.IN)",
    color = "País"
  ) +
  theme_minimal()

print(grafico)
```




```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-03 09:58:32 -03"
```